

Your grade: 90%

Your latest: 90% • Your highest: 90% • To pass you need at least 70%. We keep your highest score.

Next item →

1.

How many binary strings of length 6 exist?

1 / 1 point

64

5!

32

6!

Nice work

Each digit can accept 2 possible values $\{0, 1\}$. So the total number of binary strings is $2^6 = 64$. Note that the leftmost value can be 0 here.

2.

There are 8 atheletes. In how many ways, they can recieve a Gold, Silver, and a Bronze medal. Note that an athlete can only recieve at most 1 medal.

1 / 1 point

336

Nice work

3.

A club has 6 members. In how many ways can they select a president and a vice-president, assuming no one can hold both positions?

1 / 1 point

36

12

15

30

Nice work

The president can be chosen in 6 ways. After selecting the president, the vice-president can be chosen from the remaining 5 members. By the product rule, the total number of ways is $6 \times 5 = 30$.

4.

Assume there are 7 benches at a park in a line. The mayor decides to color these benches with 3 different colors such that no two adjacent benches have the same color. In how many ways can he do it?

1 / 1 point

192

210

729

128

Nice work

Let's give indexes to the benches from left $b_1, b_2, \dots, b_7$. We can color b_1 in 3 different ways. b_2 can only take two colors as its color must be different from b_1 . Similarly b_3 can only take 2 colors as its color must be different from b_2 . Using the same argument we can see that the total number of ways to color all these benches with this limitation is equal to $3 \times 2 \times 2 \times \dots \times 2 = 3 \times 2^6 = 192$

5.

Assume there are 4 distinct balls and 3 distinct baskets. In how many ways can we put these ball in the baskets?

1 / 1 point

4!

3⁴

4³

3!

Nice work

Each ball can go to 3 different baskets. This means each ball has 3 options and there are 4 balls. So the answer is 3^4

6.

There are 10 balls, and 3 boxes are available to store them. If each ball is placed in one box, what is the minimum number of balls in the box with the most balls?

1 / 1 point

2

3

4

5

Nice work

By the pigeonhole principle, if 10 balls (pigeons) are distributed into 3 boxes (pigeonholes), the minimum number of balls in the box with the most balls is $\lceil \frac{10}{3} \rceil = 4$.

7.

Bob is planning to attend a party. He has 4 different suits and 3 different ties. In how many ways he can choose a suit and a tie?

1 / 1 point

8

16

7

12

Nice work

He can get ready in $4 \times 3 = 12$ different ways

8.

Consider the following set $\{1, 2, \dots, 23\}$. What is the minimum number of distinct elements needed to take from this set such that the sum of two of them is 24?

0 / 1 point

13

12

11

14

Try again

9.

Using the following digits $\{0, 1, 2, 3, 4, 5\}$, how many 4-digit numbers can be formed such that are multiples of 5 if repetition is not allowed? Note that the leftmost digit can not be 0.

1 / 1 point

120

96

100

108

Nice work

For a number to be a multiple of 5, its rightmost digit must be 0 or 5. We consider two cases, with no digit repetition allowed.

Case 1: Rightmost digit is 5. The leftmost digit has 4 choices $(1, 2, 3, 4)$, as 0 and 5 are not possible, and the middle two digits have 4 and 3 choices, respectively: $4 \times 4 \times 3 = 48$.

Case 2: Rightmost digit is 0. The leftmost digit has 5 choices, and the middle two digits have 4 and 3 choices: $5 \times 4 \times 3 = 60$.

So the final answer is $60 + 48 = 108$

10.

A student has 5 different math books and 4 different physics books. In how many ways can he choose one math book and one physics book to study?

1 / 1 point

20

9

25

16

Nice work

By the product rule, the number of ways to choose one math book is 5, and one physics book is 4. Thus, the total number of ways is $5 \times 4 = 20$.